Possible Causes and Recommended Solutions for Tasks

1. Missing Value Analysis

Problem

Some columns, such as Supplier, Unit_Price, and Stock, contain missing values.

Possible Causes

- Employees may have skipped fields while entering product details.
- Supplier information may not have been available during data collection.
- Inventory and pricing systems may not have synchronized correctly.
- Data imports from multiple systems may have resulted in incomplete records.

Business Impact

- Missing unit prices can lead to inaccurate revenue and sales calculations.
- Missing stock quantities make inventory planning less reliable.
- Missing supplier details can delay purchasing and supplier follow-ups.
- Incomplete data can reduce the reliability of business reports.

Recommended Solutions

- Validate mandatory fields before saving product records.
- Retrieve missing prices from supplier invoices or purchase orders.
- Recover stock quantities from the inventory management system.
- Use fillna("Unknown") for categorical information when the actual value cannot be recovered.
- Retain unresolved numerical values as missing rather than inserting arbitrary values.
- Monitor missing-value counts and percentages regularly using Pandas.

2. Duplicate Record Detection

Problem

The dataset contains an identical product record with Product_ID 201 appearing more than once.

Possible Causes

- The same dataset may have been imported multiple times.
- Employees may have entered the same product information more than once.
- Automated data integration processes may have inserted duplicate records.
- The system may not enforce uniqueness for product identifiers.

Business Impact

- Duplicate products can inflate product counts and sales totals.
- Inventory reports may become inaccurate.
- Purchasing teams may mistakenly believe that more unique products are available.
- Business decisions based on duplicated information may be misleading.

Recommended Solutions

- Detect exact duplicates using duplicated().
- Display duplicate records before removing them.
- Remove verified duplicate rows using drop_duplicates().
- Enforce unique product IDs in the inventory database.
- Use import validation and unique constraints to prevent repeated records.
- Distinguish duplicate product master records from legitimate repeated transactions.

3. Group-Wise Analysis

Problem

The dataset contains several product categories, including Consumables, Diagnostic, Monitoring, Mobility, Emergency, and Respiratory. Their total sales values and average unit prices differ.

Possible Causes

- Different categories may have different demand levels.
- Medical equipment can cost more than consumable supplies.
- Some categories may have more products or higher sales volumes.
- Stock availability may limit the number of units sold.
- Supplier costs and product specifications may vary across categories.

Business Impact

- High-performing categories may require additional inventory.
- Low sales may indicate low demand, limited availability, or an insufficient product range.
- Differences in average prices can influence purchasing budgets.
- Incorrect or missing prices may distort category-level sales comparisons.

Recommended Solutions

- Use groupby() to calculate total units sold, average unit price, and total sales value.
- Compare categories using multiple metrics instead of relying on one measure.
- Investigate low-performing categories before reducing their inventory.
- Ensure that missing prices are verified before finalizing sales totals.

- Allocate purchasing budgets based on demand, stock levels, and business priorities.
- Review category performance regularly to identify changing trends.

4. Highest-Priced Item Analysis

Problem

The dataset contains an unusually high unit price of 999,999 for an oxygen cylinder.

Possible Causes

- An extra digit may have been entered accidentally.
- A currency conversion or decimal-formatting error may have occurred.
- The price may have been imported from an incorrect field.
- The item may have an unusual supplier quotation or specialized configuration.

Business Impact

- An incorrect price can distort average prices and descriptive statistics.
- Inventory valuation may be overstated.
- Purchasing budgets and supplier comparisons may become unreliable.
- Reports may incorrectly identify the product as exceptionally expensive.

Recommended Solutions

- Identify the maximum price using max().
- Retrieve the associated product using idxmax() and loc[].
- Display the five highest-priced items using nlargest().
- Detect potential outliers using the Interquartile Range method.
- Verify unusual prices against supplier quotations, purchase orders, and invoices.
- Correct confirmed data-entry errors using verified values.
- Retain legitimate high prices and document their reasons.
- Avoid automatically deleting an item merely because its price is an outlier.